Cross-Domain Limits of Hand-Crafted CoT-Surface Features

Strong Performance in Math, Narrower Gains in Science, Weak Results in Coding

Yuhan Chi
Fudan University
yhchi25@m.fudan.edu.cn

Abstract

Can simple summaries of chain-of-thought (CoT) traces indicate whether a solution is correct? We study one family of such features—hand-crafted CoT-surface and token-trajectory descriptors—using **DeepSeek-R1-0528-Qwen3-8B** across math, science, and coding tasks.

The results vary sharply by domain. In math, the signal is strong: our probe achieves AUC-of-AUROC (AoA) 0.958 and AUROC@100% 0.982, with dense-anchor curves plateauing early and best-of-$N = 64$ reranking improving by +10.0 percentage points. Science shows a weaker, narrower pattern: the full probe reaches AoA 0.799 and AUROC@100% 0.841, but does not exceed a simple token-confidence baseline (0.809 / 0.856), with permutation importance concentrated on confidence-based features.

Coding tells a different story. On LiveCodeBench-v5, the main probe drops to AoA 0.434 and AUROC@100% 0.407. Sweeping 83 coding-specific scalars peaks at AUROC 0.556. Grouped feature ablations remain near chance. A coding-specific CoT-only judge using 19 surface features achieves 5-fold GroupKFold out-of-fold AUC of 0.481 ± 0.053 and no best-of-64 gain (-0.006). Neither nonlinear models nor self-supervised pre-training substantially improve performance, and token-level de-knotting shows no benefit.

We therefore draw a narrower conclusion than earlier drafts: in this single-model study, hand-crafted CoT-surface features work well for math, partially for science, and poorly for coding correctness on unseen problems. This does not imply that raw-text classifiers, code-aware models, execution-aware selectors, or hidden-state probes must fail. Rather, it identifies one specific feature family whose semantics do not transfer cleanly to code verification.

1 Introduction

Process supervision and verifier-based reranking typically assume that answer quality leaves detectable traces in the generated chain of thought. This assumption often holds in mathematical reasoning, but its validity in coding remains unclear.

We examine a focused question: rather than asking whether all text-based verifiers fail on code, we investigate whether one particular family—cheap, interpretable CoT-surface features—transfers across domains. This family comprises token-confidence summaries, trajectory statistics, and activation-based descriptors. These features are computationally lightweight and straightforward to evaluate at test time.

Domain-specific results differ markedly. Mathematical reasoning exhibits strong early signals. Science problems show usable but narrower signals, primarily driven by token confidence. Coding presents a clear contrast: across multiple approaches—the main probe, extended scalar sweeps, grouped feature ablations, a coding-specific judge, nonlinear models, self-supervised pre-training, and token-level de-knotting—performance on unseen problems remains weak.

Our contribution is empirical: we demonstrate where this feature family excels (math), where it provides partial but limited utility (science), and where it fails to transfer (coding). This negative result has value—if a simple verifier does not work in a domain, it indicates that the chosen features do not reliably track correctness in that setting.

2 Related Work

Process supervision and verifiers. Outcome verifiers and process reward models established that reasoning traces can be useful supervision signals [2–4]. ProcessBench evaluates step-level error detection in mathematical reasoning [5], and generative verifier formulations cast reward modeling as next-token prediction [6]. Our paper asks a different question: before training a sophisticated verifier, what can be recovered from a very cheap feature family?

CoT faithfulness and self-correction. CoT traces are not always faithful explanations of model reasoning [7]. Related work on self-correction shows that reflective text does not reliably translate into better reasoning on its own [8–11]. We add a cross-domain angle: even when

self-corrective language is visible, the *measurability* of correctness from that language depends on the task.

Code selection and execution-aware verification. Code-generation systems often rely on information channels stronger than CoT surface form. CodeT, for example, selects candidates by executing them against generated tests and comparing execution agreement [12]. Our setting is intentionally weaker: no execution feedback, no hidden-state probes, and no external API judge. The question is whether cheap hand-crafted CoT-surface features are already enough.

Measurement invariance. The psychometric literature on measurement invariance asks whether the same instrument measures the same latent construct across groups [13]. Probing work in NLP makes a related point: a probe can be informative without being semantically stable across settings [14]. We use that lens here. The same feature name can measure "failure to converge" in math and something much less useful in coding.

3 Setup

3.1 Model, domains, and data

All traces are generated by **DeepSeek-R1-0528-Qwen3-8B** at temperature 1.0. **Math:** 7,680 aligned runs across AIME24, AIME25, BRUMO25, and HMMT25. **Science:** 12,672 runs from GPQA. **Coding:** 10,688 runs from LiveCodeBench-v5.

3.2 Feature family

The main probe uses five hand-crafted features computed from cached token streams and sparse activations.

Token-level summaries. `tok_conf` is the mean negative log probability over visible prefix tokens. Four additional per-token arrays are available for auxiliary checks: `tok_gini`, `tok_logprob`, `tok_neg_entropy`, and `tok_selfcert`.

Trajectory statistics. `traj_continuity` is the mean Jaccard similarity between consecutive activation slices. `traj_novelty` is the mean $1 - \max_j \text{Jaccard}(s_i, s_j)$ over prior slices. `traj_reflection_count` is the negated count of non-adjacent slice revisits above a fixed overlap threshold.

Neuron summary. `nc_mean` is the mean row width of the sparse activation bank over the visible prefix.

3.3 Probe pipeline and metrics

The base probe is `StandardScaler` → `TruncatedSVD` → `LogisticRegression`, fit separately per domain. Evaluation uses problem-grouped splits.

Domain	AoA	A100	Fwd Δ	Bwd Δ
Math	**.958** [.931, .980]	**.982**	−.001	−.006
Science	**.799** [.775, .822]	**.841**	−.010	−.001
Coding	.434 [.404, .464]	.407	—	—
Baselines				
`tok_conf`	.506 / .809 / .506	.490 / .856 / .490	—	—
SC oracle†	.953 / .921 / .961	.953 / .921 / .961	—	—

Table 1: Main holdout results. AoA = AUC-of-AUROC; values in brackets are 95% CIs from per-dataset bootstrap (B = 10,000). A100 = AUROC at the 100% anchor. Frozen-basis ΔAUROC: Fwd = source-to-target transfer gap; Bwd = target-to-source. Baseline rows show math/science/coding values. †SC oracle = self-consistency upper bound using ground truth.

The main early-stop summary is **AUC-of-AUROC (AoA)** over anchor positions $\{10, 40, 70, 100\}\%$. We use AoA because the paper examines how signal emerges over the trace, not only at the end. To avoid over-relying on a single summary, we also report AUROC@100% and use a denser 10-anchor check for robustness. Bootstrap 95% confidence intervals on AoA use $B = 10{,}000$ resamplings within each domain.

4 Mathematical Reasoning: Strong Signal; Science: Narrower Signal

Figure 1 and Table 1 present the main cross-domain results. Math represents the clearest success case. Science shows positive but less clean results. Coding represents the failure case.

Math: strong and structured signal. The math probe substantially outperforms the token-confidence baseline: AoA 0.958 versus 0.685, and AUROC@100% 0.982 versus 0.684. This signal emerges early. On a denser 10-anchor grid, math reaches 95% of its final AUROC by the 10% anchor and plateaus by 50%, achieving dense-anchor AoA 0.968.

Science: usable but narrower than math. Science results stay above chance and improve with more context, yet differ qualitatively from math. The full probe achieves AoA 0.799 and AUROC@100% 0.841, while a simple `tok_conf` baseline reaches slightly higher values (AoA 0.809, AUROC@100% 0.856). The science signal is

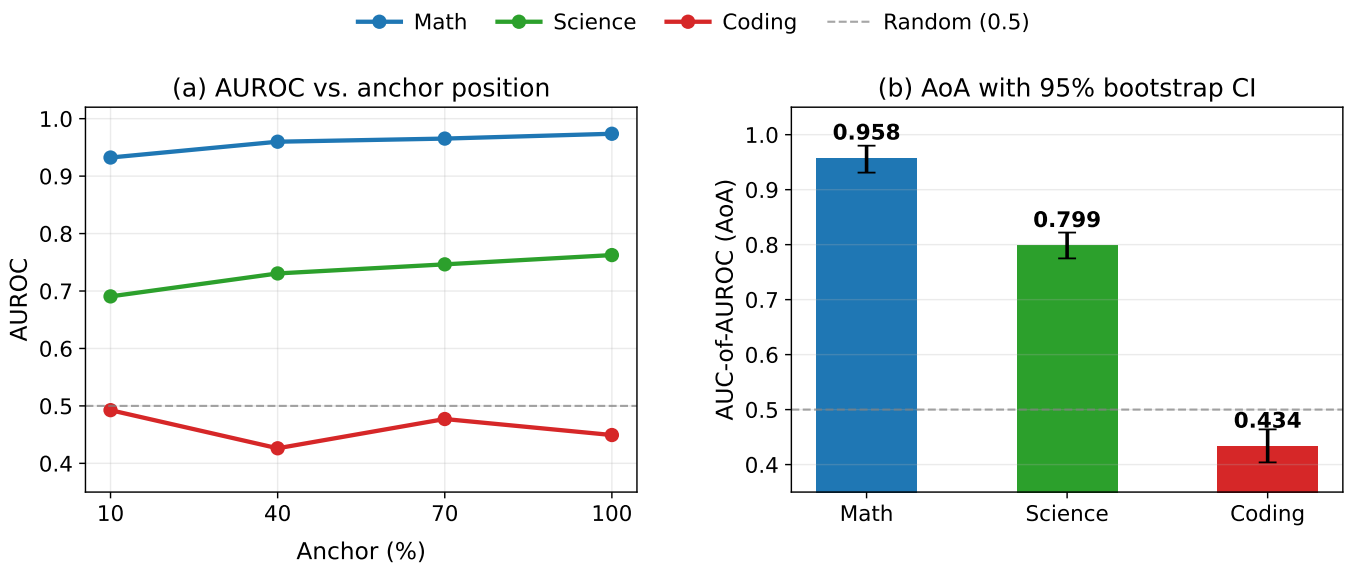

Figure 1: **(a)** AUROC at each anchor position (10%–100% of trace visible) across domains. Mathematical reasoning rises early and stays high; science improves gradually; coding remains flat. **(b)** AoA with 95% bootstrap CIs. The math–coding gap is substantial; science occupies an intermediate position.

genuine but appears largely confidence-driven rather than trajectory-structured.

Feature-importance contrast. Permutation importance highlights this difference. In math, `traj_reflection_count` dominates (importance 0.242), with `traj_continuity` (0.092) and `traj_novelty` (0.052) following—a trajectory-driven pattern. In science, the leading features are `tok_gini_tail` (0.038), `tok_conf_recency` (0.033), `tok_conf_prefix` (0.033), and then `traj_continuity` (0.031), reflecting a confidence-heavy rather than trajectory-structured signal.

Frozen-basis transfer. Cross-anchor transfer behaves differently across domains. Math exhibits near-lossless forward and backward transfer (-0.001 and -0.006), while science shows a larger forward gap (-0.010). This suggests math has a more stable quality geometry across trace positions.

Dense-anchor robustness. The four-anchor summary captures the main pattern. On a denser 10-anchor grid, the domain ordering remains unchanged: AoA reaches 0.968 for math, 0.827 for science, and 0.432 for coding. Science continues improving into later anchors, whereas coding stays weak throughout.

5 Coding: Persistent Weakness Across CoT-Surface Variants

For coding, these hand-crafted CoT-surface features fail to predict correctness on unseen problems. We demonstrate this through multiple grouped cross-validation evaluations.

5.1 Main probe and scalar screen

The main coding probe achieves AoA 0.434 and AUROC@100% 0.407, below the `tok_conf` baseline (AoA 0.506, AUROC@100% 0.490). Screening 83 coding-specific scalars yields limited improvement: the best single feature, `struct_density`, reaches oriented AUROC 0.565, followed by `cf_n_ifs` (0.563) and `cot_trace_count` (0.558). Even the optimal top-4 combination only achieves AUROC 0.556.

Interpretation. These features are not pure noise—they exhibit weak but detectable correlations. However, the strongest effects remain small and semantically mixed. For instance, deeper branching and more `if`-statements weakly correlate positively with correctness, while denser structural narration shows weak negative correlation. While variation exists, it does not reliably separate correct from incorrect solutions.

5.2 Grouped family ablations

To test whether failure stems from a specific subset rather than the entire family, we evaluate grouped feature subsets

using 5-fold GroupKFold on the 10,688 coding runs. No subset performs well: `traj_only` achieves AUROC 0.509, `token_plus_traj` reaches 0.501, and the full 30-feature surface family tops out at 0.506 in its best configuration. All remain in weak discrimination regimes.

5.3 Coding-specific CoT-only run judge

We then construct a coding-specific CoT-only judge using 19 surface features: planning and execution density, vocabulary overlap, structural flags, length summaries, one neuron count, and four one-hot `cot_mode` indicators. This model remains cheap and interpretable, evaluated on unseen `problem_id`s using 5-fold GroupKFold.

Performance remains weak. Mean out-of-fold AUC is 0.481 ± 0.053. By difficulty band, results stay near chance: hard 0.515, mid 0.470, easy 0.509. In best-of-64 reranking, the judge achieves pass@1 0.581 against a random baseline of 0.587 ($\Delta = -0.006$). Surface cues that appear meaningful within problems do not generalize to unseen problems.

5.4 Nonlinear and self-supervised checks

A one-hidden-layer MLP on the 25-dimensional feature family produces similar results: mean-anchor AUROC 0.499 and AUROC@100% 0.507.

Self-supervised pre-training provides further evidence. Cross-domain SSL v1 collapses at one rank without meaningful coding improvement. Domain-specific SSL v2 avoids collapse and offers modest low-label regularization, yet the full-label coding ceiling remains weak: the best v2 configuration achieves AUROC 0.454 at label fraction 1.0, peaking at 0.537 when label fraction is 0.05. We interpret SSL as indicating that these objectives do not uncover additional label-relevant structure in this feature space, not as proof that no useful structure exists.

5.5 De-knotting remains neutral in coding

One alternative explanation posits that coding traces contain noisy circular spans that obscure an otherwise useful signal. We test this by removing GLM-labeled knot spans from all cached per-token arrays, not just from the text.

Results differ by domain. In math, removing knot tokens degrades AUROC ($0.502 \rightarrow 0.453$, $\Delta = -0.049$), indicating these spans contribute to the signal. Science shows minimal change ($\Delta = -0.006$). Coding remains neutral ($0.460 \rightarrow 0.466$, $\Delta = +0.006$). De-knotting does not rescue coding performance.

Check	Best coding result	Reading
Main probe	AoA .434 / A100 .407	Below `tok_conf`
83-scalar screen	AUROC .556	Weak proxy only
Family ablations	AUROC .509 (`traj_only`)	No stable subset
CoT-only run judge	OOF AUC .481 ± .053	Fails on unseen problems
MLP	AUROC .499–.507	No meaningful gain
SSL v2	.537 at lf=.05; .454 at lf=1.0	Low-label regularizer only
De-knotting	$\Delta = +.006$	Neutral

Table 2: Coding-side robustness checks. Results are drawn from grouped or held-out evaluation in the artifact tables. No approach produces a strong, generalizable coding verifier using only cheap CoT-surface features.

Domain	n	Random	Probe	Δ
Math	120	.642 [.572, .712]	.742 [.658, .817]	**+.100**
Science	198	.602 [.550, .652]	.682 [.616, .748]	**+.080**
Coding	167	.617 [.545, .689]	.611 [.539, .683]	−.006

Table 3: Best-of-$N = 64$ reranking pass@1. Values in brackets are 95% bootstrap CIs from problem-level resampling.

6 Downstream Impact: Best-of-N Reranking

The cross-domain probe results affect more than just metrics—they change downstream selection behavior.

Math and science show substantial gains; coding does not. The coding-specific CoT-only judge performs similarly: out-of-fold reranking yields 0.581 against the same 0.587 random baseline. For this feature family, weak run-level discrimination translates to no useful selection improvement.

7 Discussion

What we claim. Within this single-model study, hand-crafted CoT-surface features are domain-specific tools: they work well for math, partially for science, and poorly for coding.

What we do not claim. We do not argue that correctness is fundamentally undetectable in text, nor that raw-text classifiers, code-aware models, execution-aware selectors, hidden-state probes, or LLM judges must fail. A

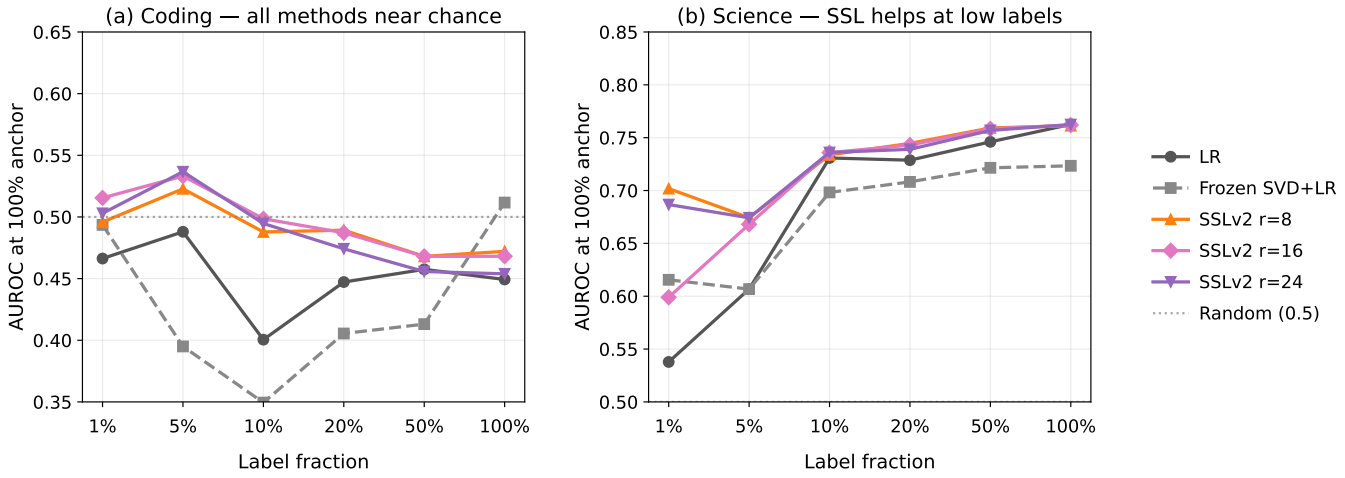

Figure 2: **SSL ceiling in coding vs. science. (a)** In coding, the full-label ceiling remains weak across SSL variants. **(b)** Science: SSL provides greater low-label benefits, consistent with a domain where features carry label-relevant structure.

more plausible explanation is that coding requires different signals, potentially execution-based ones.

Why math and coding diverge. In math, visible hesitation and convergence patterns often reflect underlying reasoning quality. Reflection loops, continuity breaks, and recency-weighted confidence track whether the reasoning path has stabilized. In coding, correctness depends on program execution rather than narrative coherence. A clean explanation can implement incorrect logic, while a messy explanation may produce correct code. The text trace is thus further removed from the target variable.

Why science occupies the middle ground. Science problems share some verbal uncertainty cues with math, but our results suggest the usable signal comes primarily from token confidence rather than strong trajectory structure. This explains why science appears positive in aggregate but weaker on closer inspection.

Future work. Subsequent work should test a second model, include raw-text or LLM-judge baselines, and add an execution-aware positive control. These experiments would clarify whether the current failure is specific to this feature family, this model, or both.

8 Limitations and Conclusion

Limitations. First, all experiments use a single model (**DeepSeek-R1-0528-Qwen3-8B**). Second, our strongest coding check remains a hand-crafted CoT-only judge; we do not include raw-text classifiers or external LLM judges. Third, the coding experiments lack an execution-aware positive control. Fourth, our SSL conclusion is objective-specific: it shows failure under the tested pretraining losses, not a universal impossibility.

Conclusion. We draw a narrower conclusion than earlier drafts. The evidence supports this statement:

> **Hand-crafted CoT-surface features transfer well to math, partly to science, and poorly to coding correctness on unseen problems in this single-model study.**

This negative result remains valuable: it shows that cheap CoT measurements should be treated as domain-specific instruments, not general-purpose correctness proxies. For coding, at least with this model and feature family, they are the wrong tool.

Code and artifacts. The public repository for this project is https://github.com/Chi-Shan0707/code-not-text. In the local snapshot used for this draft, code and paper assets remain under `workshop/cotknot/`. Paper sources are in `workshop/cotknot/paper/`. Quantitative tables used here come from `results/tables/` and `workshop/cotknot/results/tables/`. The coding-side robustness checks in this revision rely in particular on four artifacts: `results/tables/coding_feature_family_ablation.csv`, `results/tables/cot_run_judge_scores.csv`, `results/tables/cot_run_judge_rerank.csv`, and `results/models/cot_run_judge.summary.json`.

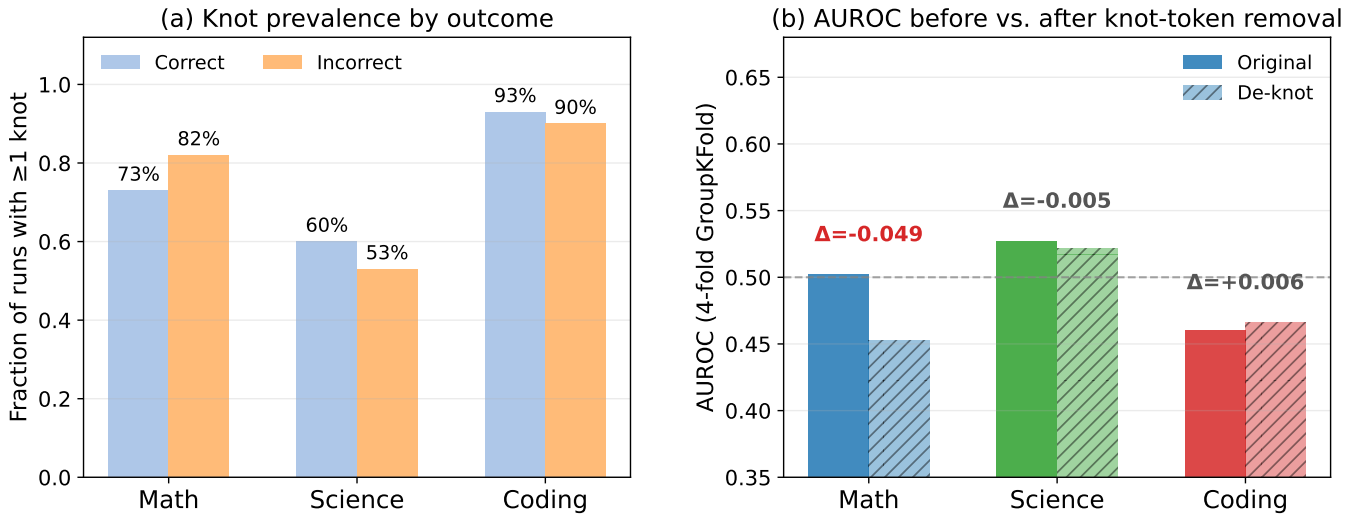

Figure 3: **De-knotting effects across domains.** Removing knot tokens degrades math performance, shows minimal effect in science, and remains neutral for coding. The coding weakness is therefore not attributable to removable surface clutter.

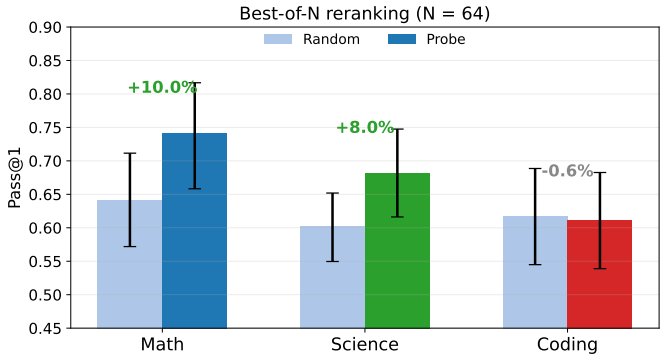

Figure 4: Best-of-$N = 64$ probe reranking results. Math gains +10.0pp, science gains +8.0pp, and coding shows no meaningful improvement.

References

[1] Jason Wei et al. Chain-of-thought prompting elicits reasoning in large language models. In *NeurIPS*, 2022.

[2] Karl Cobbe et al. Training verifiers to solve math word problems. *arXiv:2110.14168*, 2021.

[3] Hunter Lightman et al. Let's verify step by step. *arXiv:2305.20050*, 2023.

[4] Peiyi Wang et al. Math-Shepherd: Verify and reinforce LLMs step-by-step without human annotations. In *ACL*, 2024.

[5] Chujie Zheng et al. ProcessBench: Identifying process errors in mathematical reasoning. In *ACL*, 2025.

[6] Lunjun Zhang et al. Generative verifiers: Reward modeling as next-token prediction. In *ICLR*, 2025.

[7] Tamera Lanham et al. Measuring faithfulness in chain-of-thought reasoning. *arXiv:2307.13702*, 2023.

[8] Jie Huang et al. Large language models cannot self-correct reasoning yet. *arXiv:2310.01798*, 2023.

[9] Dancheng Liu et al. Large language models have intrinsic self-correction ability. In *NeurIPS*, 2024.

[10] Aman Madaan et al. Self-Refine: Iterative refinement with self-feedback. *arXiv:2303.17651*, 2023.

[11] Noah Shinn et al. Reflexion: Language agents with verbal reinforcement learning. *arXiv:2303.11366*, 2023.

[12] Bei Chen et al. CodeT: Code generation with generated tests. *arXiv:2207.10397*, 2022.

[13] Robert E. Vandenberg and Charles E. Lance. A review and synthesis of the measurement invariance literature. *Organizational Research Methods*, 3(1):4–70, 2000.

[14] Yonatan Belinkov. Probing classifiers: Promises, shortcomings, and advances. *Computational Linguistics*, 48(1):207–219, 2022.